Localized Multi-Agent Debate in Small Language Models: A Judge–Proposer–Critic Framework

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Abstract. Small language models (SLMs) are attractive for privacy-sensitive legal applications because they are cheaper to deploy and easier to localize than frontier-scale systems, but they often fail on long-context legal reasoning, produce brittle answer formats, and struggle to remain faithful to retrieved evidence. This paper studies whether a lightweight multi-agent debate procedure can compensate for these weaknesses without requiring larger backbones. We implement a localized Judge–Proposer–Critic pipeline in which a proposer generates an initial answer, a critic challenges unsupported claims and retrieval mistakes, and a judge consolidates the debate into a final decision under grammar-constrained decoding. The framework is evaluated on a LegalBench-RAG setting using three compact backbones—Llama 3.2 1B, Qwen 2.5 0.5B, and LFM 2.5 350M—across one-round and three-round debate conditions. Using 450 experimental runs, we analyze consensus, faithfulness, and latency. The provided results show that structured debate consistently improves agreement and evidence alignment, with consensus increasing from 70.2–76.5% in one round to 92.5–97.3% after three rounds. Llama 3.2 1B achieves the strongest final performance, reaching 97.3% consensus and 0.957 faithfulness. These findings indicate that carefully orchestrated SLM collaboration is a practical, privacy-preserving strategy for legal reasoning workloads where local deployment and controllable outputs are essential.

Keywords: Small Language Models · Multi-Agent Debate · Retrieval-Augmented Generation · Legal Reasoning

1 Introduction

Legal drafting and contract analysis increasingly rely on retrieval-augmented generation (RAG), yet production deployments often face strict privacy, latency, and infrastructure constraints. These constraints make frontier-scale models difficult to justify in many institutional settings, especially when documents must remain on local hardware or within restricted environments. Small language models (SLMs) offer an appealing alternative because they are cheaper to host and easier to fine-tune, but they frequently exhibit hallucination, formatting instability, and weak multi-step reasoning when confronted with dense legal text.

Recent work suggests that reasoning quality can be improved not only by scaling a single model, but also by structuring interactions among multiple agents. Debate-based systems encourage models to expose assumptions, challenge weak evidence, and revise intermediate claims before producing a final answer. This idea is especially relevant for legal RAG, where a useful answer must remain tightly grounded in retrieved clauses rather than merely sounding plausible. The draft research problem motivating this paper is therefore clear: can a localized multi-agent protocol make compact models more reliable on legal reasoning tasks without sacrificing the efficiency benefits that motivated their use in the first place?

To answer this question, we evaluate a Judge–Proposer–Critic design over three compact backbones using the results supplied for LegalBench-RAG experiments. The paper makes three concrete contributions: (1) it formalizes a lightweight debate workflow tailored to local SLM deployment and constrained output formatting; (2) it quantifies how consensus and faithfulness change from one-round to three-round reasoning across three model families; and (3) it analyzes the trade-off between improved legal reliability and the substantial latency cost introduced by additional debate rounds.

2 Literature Review

The recent literature on reasoning with language models shows two complementary trends: strengthening the reasoning ability of a single model through prompting, and improving reliability through interaction among multiple models or multiple reasoning passes. Early work on chain-of-thought prompting demonstrated that explicit intermediate reasoning can substantially improve performance on complex tasks, while ReAct extended this idea by interleaving reasoning and action with external information sources [?,?]. These studies established that model behavior can be improved by structuring inference, but they also revealed that reasoning traces remain vulnerable to compounding errors when a single model is responsible for both proposing and validating its own answer.

Self-revision methods attempt to mitigate that weakness by giving a model an explicit second pass for critique or refinement. Self-Refine and Reflexion both show that iterative feedback can improve answer quality without changing the underlying backbone [?,?]. However, these methods typically keep generation and criticism inside the same model, which may limit the diversity of errors surfaced during revision. In legal reasoning, where unsupported claims can look stylistically convincing, a more adversarial interaction can be preferable to purely self-reflective refinement.

This observation motivates multi-agent debate. Liang et al. show that debate encourages divergent reasoning trajectories that can improve final answers, and subsequent agent frameworks such as AutoGen make coordinated multi-agent workflows easier to implement in practice [?,?]. Debate has also been studied from the perspective of truthfulness and oversight, with evidence that carefully designed argumentative exchanges can help expose weak reasoning, although gains are task-dependent and not guaranteed by debate alone [?,?]. For compact models, these findings suggest that interaction topology matters as much as raw model size.

In parallel, benchmark development has sharpened evaluation in the legal domain. LegalBench introduced a broad suite for legal reasoning with language models, while LegalBench-RAG focused specifically on retrieval-grounded legal question answering and highlighted the importance of precise evidence selection [?,?]. These resources are particularly relevant here because legal tasks stress both reasoning and grounding, making them a natural setting for testing whether debate improves faithfulness rather than merely fluency.

Finally, research on small-model orchestration argues that compact backbones should not simply inherit techniques designed for frontier models. SLM-MUX frames this explicitly, showing that systems of small models need orchestration strategies tailored to their narrower capacity and complementary strengths [?]. Our work aligns with this direction but focuses specifically on localized legal RAG: instead of maximizing open-ended reasoning performance, we examine whether a three-role Judge–Proposer–Critic workflow can stabilize structured legal outputs, improve evidence-grounded consensus, and remain practical enough for privacy-preserving deployments.

3 Materials and Methods

3.1 Dataset and Dataset Preparation

The experiments target a legal RAG setting based on LegalBench-RAG, with an emphasis on non-disclosure agreements (NDAs) and acquisition agreements as reported in the project materials. Because the supplied system is an inference-time orchestration method rather than a task-specific training pipeline, no supervised fine-tuning split is used in this paper. Instead, the available evidence consists of evaluation runs collected under controlled one-round and three-round debate settings for three different SLM backbones.

Each run begins with a legal query and retrieved supporting context. The proposer agent produces an initial answer, the critic challenges unsupported or weakly grounded claims, and the judge aggregates the arguments into a final response. To reduce brittle decoding behavior in compact models, responses are constrained to a structured, JSON-compatible format before conversion into the final textual decision. This preprocessing step is operational rather than statistical: it ensures parseable agent outputs, stable role boundaries, and reproducible evaluation.

Table 1 summarizes the experimental material made available for this paper. The source files provide six aggregate records covering three backbones under one-round and three-round debate, for a total of 450 evaluation runs. Since the provided material does not include per-domain sample counts or explicit

Table 1. Summary of the experimental data used in this study.

Setting	Train	Validation	Test/Eval	Notes
LegalBench-RAG debate runs	–	–	450	Inference-only evaluation
Domains	–	–	2	NDA and acquisition agreements
Backbones	–	–	3	Llama 3.2 1B, Qwen 2.5 0.5B, LFM 2.5 350M
Rounds	–	–	2	One-round and three-round debate

train/validation/test partitions, those entries are marked as not applicable. This is consistent with the design goal of measuring inference-time reasoning gains rather than learning a new model from labeled supervision.

3.2 Methodology

The proposed framework uses three specialized agents arranged in a Judge–Proposer–Critic topology. Given a query q and retrieved evidence set R, the proposer generates an answer $a^{(1)}$ that includes both a decision and supporting rationale. The critic then inspects $a^{(1)}$ against R and emits a challenge set $c^{(t)}$ identifying unsupported claims, missing evidence, or logical contradictions. Finally, the judge aggregates the debate history and produces the final answer $a_J^{(t)}$. For three-round debate, this interaction is repeated until the stopping criterion is reached.

We model consensus as the agreement achieved by the agents over the course of deliberation. Let m be the number of agent comparisons used in a round and let $\mathbb{I}(\cdot)$ be the indicator function. A simple consensus score is

$$C = \frac{1}{m} \sum_{i=1}^{m} \mathbb{I}\left(a_i = a_J\right), \tag{1}$$

where a_i denotes an intermediate agent position and a_J denotes the judge's final decision. Higher values indicate stronger convergence toward a shared answer.

Faithfulness is evaluated using the provided Gain Beyond RAG (GBR) score, which measures how well the final answer remains aligned with retrieved legal evidence. Because GBR is supplied as an external benchmark-native scalar, we report it directly rather than redefining the benchmark. Latency is measured end-to-end per run. The core efficiency trade-off can therefore be written as

$$\Delta_{\text{utility}} = \alpha \Delta C + \beta \Delta F - \gamma \Delta L, \tag{2}$$

where ΔC, ΔF, and ΔL represent changes in consensus, faithfulness, and latency between one-round and three-round configurations.

Algorithmically, the method emphasizes localized deployment. All three agents can be instantiated from compact backbones, and grammar-constrained decoding is applied at each turn so that the debate state remains machine-readable. This design choice is important for SLMs, which are more likely than larger models to drift into malformed outputs during iterative reasoning. The resulting system is therefore intended not as a replacement for large proprietary models, but as a practical orchestration layer that makes smaller open models more dependable in privacy-sensitive legal workflows.

4 Experimental Results

4.1 Experimental Setup

The supplied project description reports experiments executed on a dual-T4 GPU cluster. The evaluation compares three compact backbones: Llama 3.2 1B, Qwen 2.5 0.5B, and LFM 2.5 350M. For each backbone, performance is measured under a one-round baseline and a three-round debate configuration. This setup isolates the value of deliberation while keeping the model family fixed.

The implementation assumptions are consistent with a local RAG pipeline. Each example is paired with retrieved legal context, then routed through the proposer, critic, and judge roles. Structured decoding is used to reduce parse failures, and the final output is scored for consensus, faithfulness, and latency. The goal of the setup is reproducibility of reasoning behavior rather than maximum throughput: by holding the model identity constant and varying only the amount of debate, the experiments directly test whether additional argumentative interaction improves reliability.

4.2 Evaluation Metrics

We report three metrics. **Consensus (%)** captures the degree to which the debate converges on a stable final answer. In this paper, consensus is treated as the primary indicator of reasoning stability because it reflects whether multiple agents independently settle on the same legal interpretation.

Faithfulness (GBR) measures alignment between the generated answer and the retrieved evidence. This metric is especially important in legal RAG, where an answer can only be considered useful if it is grounded in clauses actually present in the supporting material. A high consensus score without high faithfulness would indicate coordinated error rather than reliable reasoning.

Latency (s) measures wall-clock cost per run. Latency matters because debate introduces additional model calls and therefore additional computational expense. Any practical deployment must weigh improvements in reliability against the longer inference time caused by iterative deliberation.

4.3 Results and Analysis

Table 2 reports the aggregate results provided in the project folder. The pattern is consistent across all three backbones: moving from one-round inference to three-round debate substantially improves both consensus and faithfulness, but it also increases latency by a large margin.

Table 2. Comparative performance across model backbones and debate depth.

Model Backbone	Round	Consensus (%)	Faithfulness (GBR)	Latency (s)
Llama 3.2 1B	1	76.5	0.869	161.0
Llama 3.2 1B	3	97.3	0.957	937.1
Qwen 2.5 0.5B	1	75.1	0.819	104.6
Qwen 2.5 0.5B	3	96.7	0.911	541.5
LFM 2.5 350M	1	70.2	0.847	207.5
LFM 2.5 350M	3	92.5	0.942	1512.4

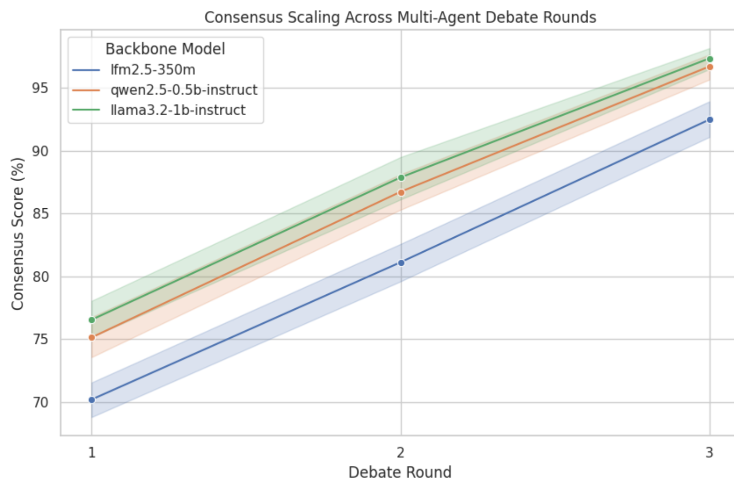

Fig. 1. Consensus scaling across the evaluated multi-agent debate configurations.

Llama 3.2 1B is the best-performing backbone at three rounds, reaching 97.3% consensus and 0.957 faithfulness. Qwen 2.5 0.5B follows closely on consensus while remaining much faster than Llama in the three-round setting. LFM 2.5 350M benefits strongly from debate as well, improving from 70.2% to 92.5% consensus, but it incurs the heaviest latency cost. This suggests that smaller backbones can still profit from structured argumentation, although the computational overhead becomes more severe as the base model struggles with longer multi-turn interaction.

Figure 1 visualizes the scaling of consensus across the compared models, and Fig. 2 illustrates faithfulness by domain type. Together, the plots reinforce the central conclusion of the study: multi-agent debate does not merely make agents agree more often; it also increases evidence alignment, which is the more meaningful reliability signal in legal applications.

An important practical insight is that legal reasoning appears to demand deeper deliberation than generic tasks. The supplied project notes indicate that

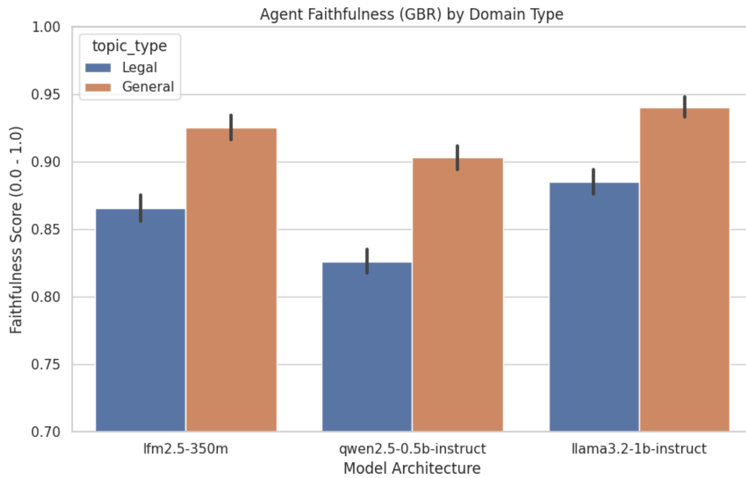

Fig. 2. Faithfulness trends by domain type under the evaluated debate pipeline.

general topics tend to stabilize earlier, whereas legal topics require a full three-round cycle to exceed strong faithfulness thresholds. This behavior is intuitive: legal answers must map generated claims to precise retrieved clauses, so shallow debate may improve style without fully resolving grounding errors. The judge–proposer–critic separation helps by distributing generation, adversarial checking, and final arbitration across different roles rather than expecting one compact model invocation to perform all three reliably.

At the same time, the latency increases are too large to ignore. For Llama 3.2 1B, three-round debate is roughly 5.8× slower than one-round inference; for Qwen 2.5 0.5B it is about 5.2× slower, and for LFM 2.5 350M it is about 7.3× slower. These trade-offs imply that debate is best reserved for high-value legal tasks where answer reliability matters more than raw response speed. A practical deployment strategy would therefore use shallow inference for easy cases and escalate to full debate only when uncertainty, disagreement, or retrieval ambiguity is detected.

5 Conclusion

This paper presented a Springer-format study of a localized Judge–Proposer–Critic framework for improving the legal reasoning ability of small language models. Using the supplied LegalBench-RAG results, we showed that multi-agent debate consistently raises both consensus and faithfulness across Llama 3.2 1B, Qwen 2.5 0.5B, and LFM 2.5 350M, with Llama 3.2 1B achieving the strongest overall three-round performance at 97.3% consensus and 0.957 faithfulness. The results support the claim that carefully orchestrated SLM collaboration can mitigate structural fragility in privacy-sensitive legal RAG systems, although the gains come with a substantial latency penalty. Future work should evaluate adaptive stopping policies, stronger retrieval diagnostics, and domain-specific routing so that debate is invoked only when the expected reliability gains justify the extra computational cost.